

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details <Do not mention any personally identifiable information like email ID, institute details, mobile phone number etc.>

Batch Number: 13

Start date: 2 February 2026

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Internship Duration: 8 Weeks

1. Project Title

Stock_Trend_AI Stock Market Forecasting Model

2. Project Objective

Build a simple, ML Model that predicts next-day closing price for Apple (AAPL) using historical data (2020–2025), validate and quantify model performance, handle data quality issues (missing values, outliers), and deliver a minimal Flask dashboard that visualizes Actual vs Predicted prices for user inspection and basic decision support

3. Project description in detail

Milestone-4 integrates and finalises work from Milestone-1 (data collection & baseline model), Milestone-2 (imputation, outlier handling, evaluation) and Milestone-3 (model training + dashboard). The final deliverable is a reproducible repo containing:

- cleaned dataset, feature engineering scripts, saved scaler, and a simple Flask app that serves:
 - Actual vs Predicted time-series plot,
 - A small table of sample Actual vs Predicted values,
 - Basic summary statistics (MAE, sample size).

Impact / real-world relevance: shows how a compact ML pipeline and dashboard can turn raw market CSVs into a repeatable deliverable for demonstrations, teaching, or prototype decision-support.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Data collection & initial inspection	Collected 5-yr AAPL CSV; removed header garbage row; stored raw copy.
Week 2	Preprocessing: types, date conversion, sort	Converted <code>Date</code> to datetime, numeric conversion for price & volume, sorted, removed invalid rows.
Week 3	Feature engineering (lags, MA), save cleaned CSV	Created <code>lag_1..lag_5</code> , <code>ma_7</code> , <code>ma_21</code> , <code>target</code> ; saved <code>dataset/apple_clean.csv</code> (1255 rows after cleaning).
Week 4	Missing value strategy + outlier handling (IQR)	Applied forward-fill for short gaps; removed extreme rows using IQR rule and recorded before/after stats.
Week 5	Model training (RandomForest baseline)	Trained Random Forest Regressor on 80% train / 20% test split; saved model and scaler.
Week 6	Dashboard (Flask) development & plotting	Built Flask app to show Actual vs Predicted plots, distribution histograms, table preview
Week 7	Integration, evaluation, diagnostic plots	Created time-series before after cleaning overlays, boxplots, histograms (saved as fig files). Prepared result comparison tables (sample rows).
Week 8	Documentation, final report & presentation	Assembled Milestone-4 report ,prepared slides/screenshots for presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Repo & initial dataset located; scope defined	12/02/2026
Prototype/First Draft	Cleaned data + feature engineering + baseline model saved	18/02/2026
Mid-Term Review	IQR outlier handling + imputation strategy validated	23/02/2026
Final Submission	Trained model + Flask dashboard + documentation	08/04/2026
Presentation	Demo of dashboard & results (Actual vs Predicted)	08/04/2026

5b. Project execution details

1. Data ingestion

- Loaded raw CSV into a pandas DataFrame; inspected head & dtypes. Removed an extra header row containing "AAPL" that broke parsing.

2. Preprocessing

- Convert Date → datetime, sort ascending, convert numeric columns (Close, High, Low, Open, Volume) using `pd.to_numeric(errors='coerce')`.
- Handle missing values: forward-fill (`df.ffill()`) for short gaps; ensured no duplicate dates.

3. Outlier detection & handling

- Applied Tukey IQR rule per numeric column and removed rows outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$. Kept a sample of removed rows for manual review (to avoid deleting genuine market events).

4. engineering

- Created lag features `lag_1..lag_5` (shifted closes), `ma_7`, `ma_21` and `target = Close.shift(-1)`. Dropped rows with NaNs from shift/rolling.

5. Train/test split and modelling

- Time-based split: `split = int((len(df)*0.8)` (80% train, 20% test). `StandardScaler` fitted on training features and saved. `RandomForestRegressor` trained and saved with `joblib`. Observed Test MAE $\approx$ **33.61** (from Milestone-1 run).

6. Evaluation & diagnostics

- Metrics: MAE, RMSE, R^2 recorded. Diagnostic plots: Actual vs Predicted line plot, histograms, boxplots, time-series overlay before/after cleaning. Saved images (e.g., `fig_time_before_after.png`, `fig_box_close.png`).

7. Dashboard

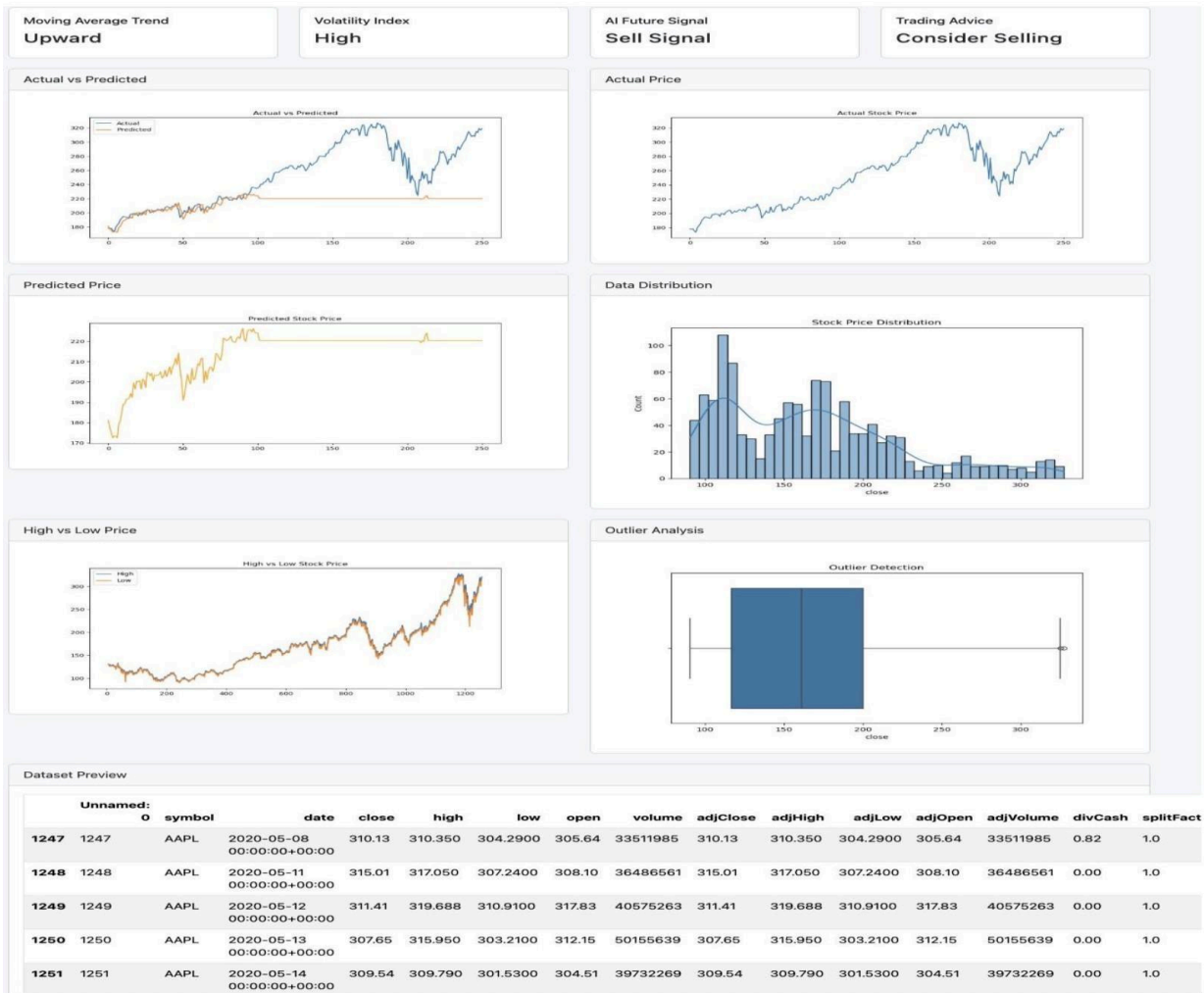
- Flask app renders:
 - Actual vs Predicted plot,
 - Sample comparison table,
 - Data distribution histogram and outlier boxplot,
 - Simple "AI signal" text box (prototype). Code follows the simple Flask structure used in Milestone-3.

8. Packaging & artifacts

• • Saved artifacts: `dataset/apple_clean.csv` (1255 rows), `models/apple_model.pkl`, `models/apple_scaler.pkl`, and scripts: `apple_preprocess.py`, `apple_model_setup.py`, `app.py` (Flask).

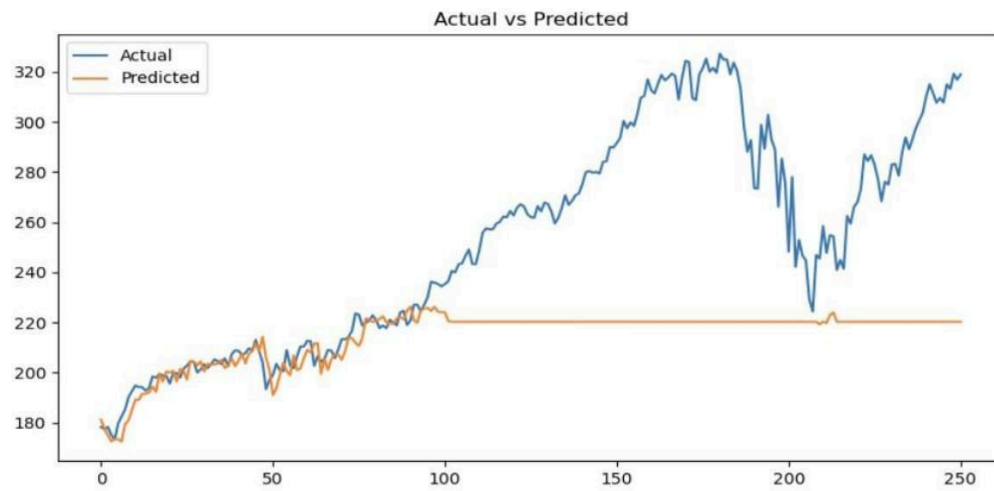
6. Snapshots / Screenshots

OVERALL DASHBOARD:

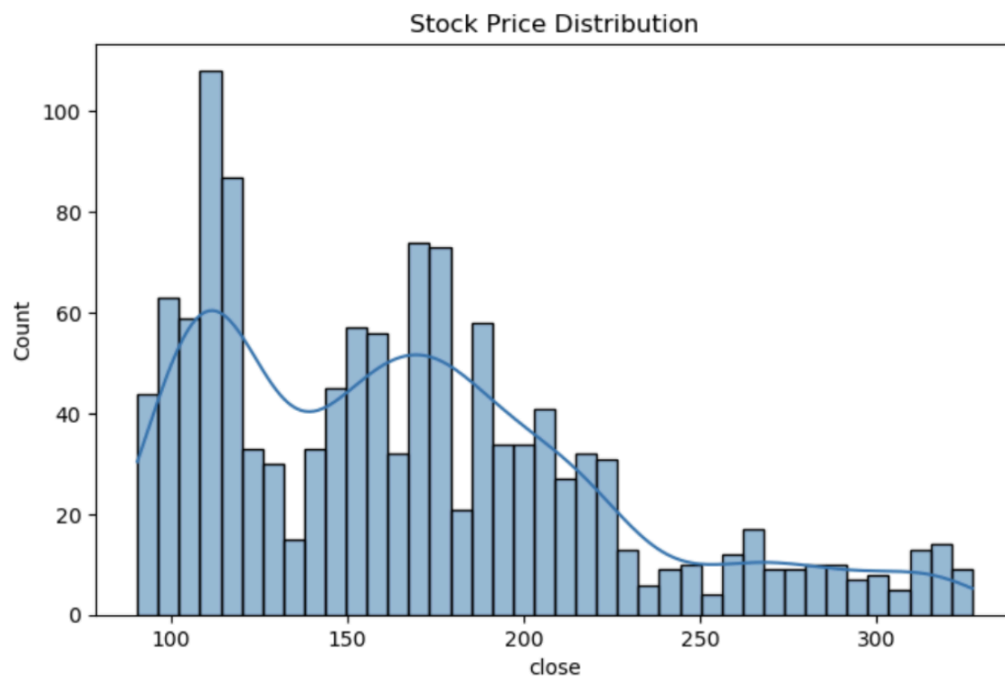


	symbol	date	close	high	low	open	volume	adjClose	adjHigh	adjLow	adjOpen	adjVolume	divCash	splitFactor
0	AAPL	2015-05-27 00:00:00+00:00	132.045	132.26	130.05	130.34	45833246	121.6825575315	121.88068506280000	119.8441183458	120.11136013220000	45833246	0.0	1.0
1	AAPL	2015-05-28 00:00:00+00:00	131.78	131.95	131.1	131.86	30733309	121.4383538301	121.5950128084	120.8117179172	121.5120757022	30733309	0.0	1.0
2	AAPL	2015-05-29 00:00:00+00:00	130.28	131.45	129.9	131.23	50884452	120.0560687281	121.13425110770000	119.70588983560000	120.9315159594	50884452	0.0	1.0
3	AAPL	2015-06-01 00:00:00+00:00	130.535	131.39	130.05	131.2	32112797	120.29105719550000	121.0789597036	119.8441183458	120.9038702574	32112797	0.0	1.0
4	AAPL	2015-06-02 00:00:00+00:00	129.96	130.655	129.32	129.86	33667627	119.7611812397	120.40164000360000	119.1714062628	119.6690288995	33667627	0.0	1.0
5	AAPL	2015-06-03 00:00:00+00:00	130.12	130.94	129.9	130.66	30983542	119.9086249839	120.664274173	119.70588983560000	120.4062476206	30983542	0.0	1.0
6	AAPL	2015-06-04 00:00:00+00:00	129.36	130.58	128.91	129.58	38450118	119.20826719890000	120.3325257485	118.79358166830000	119.4110023472	38450118	0.0	1.0
7	AAPL	2015-06-05 00:00:00+00:00	128.65	129.69	128.36	129.5	35626800	118.55398558390000	119.51236992130000	118.2867437975	119.3372804751	35626800	0.0	1.0
8	AAPL	2015-06-08 00:00:00+00:00	127.8	129.21	126.83	128.9	52674786	117.7706908928	119.07003868870000	116.87681299340000	118.7843664342	52674786	0.0	1.0
9	AAPL	2015-06-09 00:00:00+00:00	127.42	128.08	125.62	126.7	56075420	117.4205118002	118.0287172451	115.7617696778	116.7570149513	56075420	0.0	1.0

Actual vs Predicted



Data Distribution



7. Challenges Faced

1. Understanding the dataset structure

Initially, it was difficult to identify which columns should be used as input features and which should be the prediction target in the stock dataset. This issue was resolved by exploring the dataset using Pandas and selecting the Close price as the prediction target.

2. Handling missing values in the dataset

Some rows contained missing or invalid values, which caused errors during model training. This problem was mitigated by applying the forward fill (ffill) method to replace missing values with the previous valid data.

3. Detecting outliers in stock data

Certain values in the dataset were extremely high or low, which could negatively affect the prediction accuracy of the model. This was resolved by using the Interquartile Range (IQR) method to detect and remove outliers.

4. Integrating the ML model with the Flask dashboard

Connecting the trained machine learning model with the Flask web dashboard was initially challenging. This was resolved by saving the trained model using Joblib and loading it into the Flask application for predictions.

8. Learnings & Skills Acquired

1. **Data preprocessing and analysis:** Learned to explore datasets using Pandas, clean data, and convert columns like Date into proper formats for analysis.
2. **Handling missing values and outliers:** Gained practical experience in identifying missing values and detecting outliers using methods like forward fill and the IQR technique.
3. **Feature engineering for time-series data:** Learned to create lag features and moving averages to help the model understand past stock trends.
4. **Machine learning model training:** Developed skills in training and evaluating a Random Forest regression model using Scikit-learn.
5. **Web integration using Flask:** Learned how to integrate the trained machine learning model into a Flask dashboard to display predictions and visualizations.

9. Testimonials from team

1. Working on this project helped us understand the practical steps involved in building a machine learning model, from data preprocessing to prediction.
2. The milestone-based approach made the project easier to manage and helped us gradually improve our understanding of data analysis and model development.
3. Integrating the machine learning model with a Flask dashboard gave us valuable experience in connecting backend models with a simple web interface.

10. Conclusion

This internship project helped in understanding the basic workflow of a machine learning application, starting from data collection and preprocessing to model training and prediction. The project also provided practical experience in handling real datasets and integrating the trained model with a Flask dashboard. Overall, it improved technical skills in data analysis, machine learning, and basic web integration while demonstrating how ML can be applied to predict stock price trends.

11. Acknowledgements

I would like to sincerely thank **Infosys Springboard** for providing the opportunity to participate in this virtual internship program. I am grateful to the mentors and instructors for their guidance, resources, and support throughout the project.

I would also like to thank my **team members** for their collaboration and teamwork during the completion of this project. Their ideas, discussions, and support helped in successfully completing each milestone of the internship.